BEYOND MULTIPLE LINEAR REGRESSION · WEEK 2

# Treatment effects across centers

## Interactions and uncertainty

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METHODS · DESIGN · STATISTICS

# Week 1 recap: comparing the models

### THE REHABILITATION TRIAL

Simulated trial: 120 patients after knee replacement at six centers.

Within each center: 10 patients randomized to each treatment.

Standard care: weekly physiotherapy and home exercises.

Enhanced care: standard care plus a weekly supervised session.

Outcome: six-minute walk distance at 12 weeks.

### THE TREATMENT COMPARISON

| Model | Estimate (m) | Estimated SE (m) |
|---|---|---|
| Treatment only | 11.0 | 5.77 |
| Fixed center effects | 11.0 | 4.34 |
| Random center effects | 11.0 | 4.34 |

All three models assumed a common treatment effect across centers.

Fixed center parameters and a random center distribution gave the same treatment estimate and estimated standard error in this setting.

# Changing treatment allocation within centers

### ORIGINAL ALLOCATION

| Center | Standard care | Enhanced care |
|---|---|---|
| A | 10 | 10 |
| B | 10 | 10 |
| C | 10 | 10 |
| D | 10 | 10 |
| E | 10 | 10 |
| F | 10 | 10 |
| Total | 60 | 60 |

### UNEQUAL ALLOCATION

| Center | Standard care | Enhanced care |
|---|---|---|
| A | 5 | 15 |
| B | 8 | 12 |
| C | 10 | 10 |
| D | 10 | 10 |
| E | 12 | 8 |
| F | 15 | 5 |
| Total | 60 | 60 |

Only allocation changes: the underlying treatment effect remains 10 m in every center.

# The models now give different treatment estimates

Unequal allocation within centers; estimated enhanced-minus-standard difference in walk distance.

| Model | Estimate (m) | Estimated SE (m) |
|---|---|---|
| **Treatment only** | 3.87 | 5.75 |
| **Fixed center effects** | 8.40 | 4.56 |
| **Random center effects** | 8.13 | 4.55 |

Accounting for center changes the estimate substantially. The two center models give slightly different estimates, with almost identical estimated standard errors.

# The models use different comparisons

### TREATMENT ONLY

The model compares the average outcome of all enhanced-care patients with that of all standard-care patients.

Unequal allocation gives the groups different mixtures of centers. Their difference reflects both treatment and differences in center baselines.

### FIXED CENTER EFFECTS

The model allows each center to have its own baseline outcome.

It estimates treatment by combining comparisons between enhanced and standard care within the same center.

These comparisons remove the shared center baseline.

### RANDOM CENTER EFFECTS

The model uses comparisons both within and between centers.

Do centers with more enhanced care have higher average outcomes?

Using between-center information assumes baseline center effects are unrelated to treatment allocation.

The random-intercept model also uses between-center information under additional assumptions, so its treatment estimate can differ.

# Fixed versus random center effects

| Design | Treatment estimates | Estimated standard errors |
|---|---|---|
| Equal center sizes;<br>50:50 allocation within each center | Identical | Identical |
| Unequal center sizes;<br>50:50 allocation within each center | Identical | Can differ |
| Treatment proportions differ across centers | Can differ | Can differ |

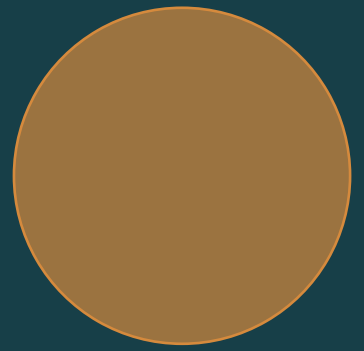
With 50:50 allocation within every center, all three models, including the treatment-only model, give the same treatment estimate, even when center sizes differ.

So far, we have assumed the same treatment effect in every center. What changes when the treatment effect varies across centers?

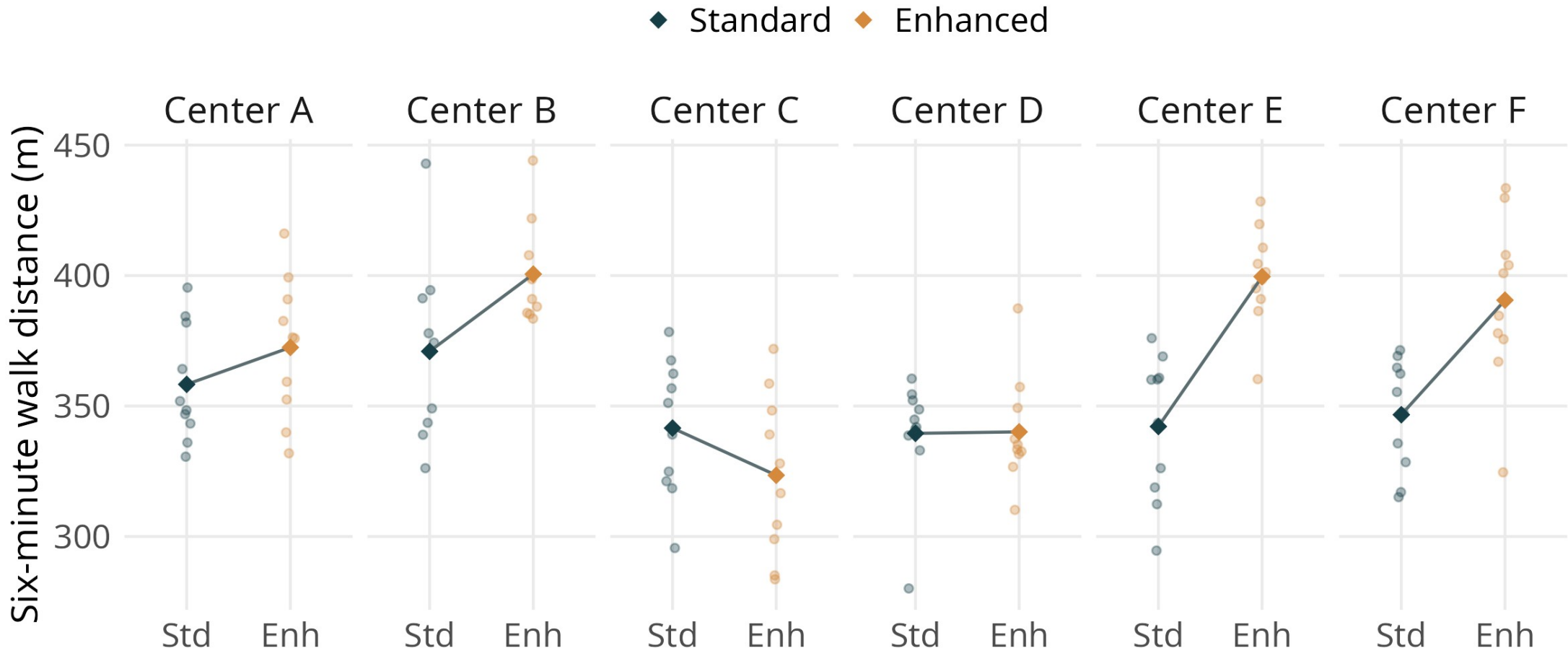
*PART 1*

# Treatment-by-center interaction





# Restore balanced allocation and let treatment effects vary




*New simulated trial: 120 patients, six centers, 10 per treatment per center. Diamonds are observed treatment means.*

# Extend the fixed-effects model

### A TREATMENT DIFFERENCE FOR EACH CENTER

$$Y_{ij} = \beta_0 + c_j + (\beta_1 + d_j)T_{ij} + \varepsilon_{ij}$$

$T_{ij}$ = 0 for standard care and 1 for enhanced care.

$$\delta_j = \mu_{1j} - \mu_{0j} = \beta_1 + d_j$$

The interaction contribution $d_j$ allows the expected treatment difference to vary by center.

### FIXED PARAMETERS AND INDIVIDUAL ERRORS

Each $c_j$ and $d_j$ is a fixed, unknown parameter for one of the six included centers.

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

After accounting for the center-specific treatment means, individual errors are independent with a common variance.

We retain the Week 1 center effects and add center-specific deviations from the average treatment effect.

# Effects coding defines the average and the deviations

### THE AVERAGE PARAMETERS

$$\beta_0 = \frac{1}{J}\sum_{j=1}^{J}\mu_{0j}$$

$\beta_0$: expected standard-care outcome, averaged equally across the included centers.

$$\beta_1 = \frac{1}{J}\sum_{j=1}^{J}\delta_j$$

$\beta_1$: treatment difference, averaged equally across those same centers.

### CENTER-SPECIFIC DEVIATIONS

$$c_j = \mu_{0j} - \beta_0, \qquad d_j = \delta_j - \beta_1$$

$c_j$ and $d_j$ describe a center's deviations in standard-care mean and treatment effect.

$$\sum_{j=1}^{J} c_j = 0, \qquad \sum_{j=1}^{J} d_j = 0$$

Center F's deviations are minus the sums of the corresponding five estimated deviations.

Treatment uses dummy coding; center uses effects coding, as in Week 1.

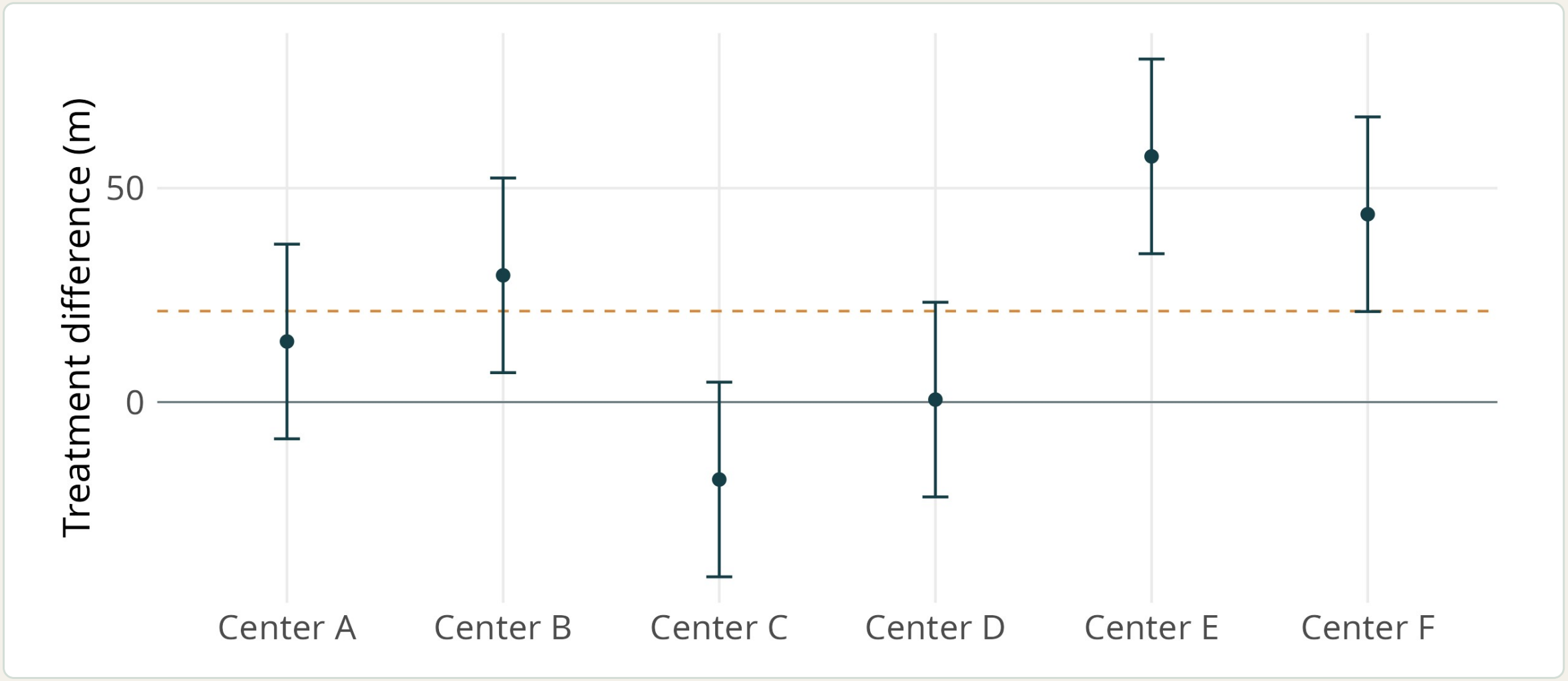
# Interpret the interaction through fitted means

Predict the mean for each treatment–center combination: the reference grid used by estimated marginal means.

| Center | Standard (m) | Enhanced (m) | Difference (m) |
|---|---|---|---|
| **Center A** | 358.3 | 372.5 | 14.2 |
| **Center B** | 370.9 | 400.6 | 29.6 |
| **Center C** | 341.6 | 323.5 | -18.1 |
| **Center D** | 339.5 | 340.1 | 0.6 |
| **Center E** | 342.2 | 399.6 | 57.4 |
| **Center F** | 346.7 | 390.6 | 43.9 |

The full interaction model reproduces the observed cell means. Subtract within each center to obtain its treatment comparison.

# Treatment differences and their uncertainty

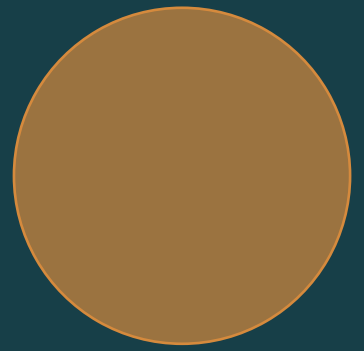



*Enhanced minus standard care. Bars: pointwise 95% confidence intervals, unadjusted. Dashed line: average estimate.*

*PART 2*

# Uncertainty in the included centers





# Which average treatment effect are we estimating?

$$\beta_1 = \frac{1}{J}\sum_{j=1}^{J}\delta_j$$

Our current target is the equally weighted average true treatment effect in these six centers.

$$\widehat{\beta}_1 = \frac{1}{J}\sum_{j=1}^{J}D_j$$

$D_j$ is the observed enhanced-minus-standard difference in center j. Here, their average is 21.27 m.

Averaging fitted means over centers and then contrasting the treatments gives the same estimate.

An average remains well defined when treatment effects differ. The weighting and the target centers must be explicit.

# Within a center, the baseline contribution cancels

$$D_j = \bar{Y}_{1j} - \bar{Y}_{0j}$$

$$\begin{aligned} D_j &= (\beta_0 + c_j + \beta_1 + d_j + \bar{\varepsilon}_{1j}) \\ &\quad - (\beta_0 + c_j + \bar{\varepsilon}_{0j}) \\ &= \beta_1 + d_j + \bar{\varepsilon}_{1j} - \bar{\varepsilon}_{0j} \end{aligned}$$

Both treatment groups share $\beta_0 + c_j$, so these terms cancel. The center-specific treatment deviation $d_j$ remains.

Each barred epsilon averages the individual errors in one treatment group within that center.

The within-center comparison removes baseline differences, but it retains differences in treatment effects.

# Uncertainty in one center’s treatment difference

### INDIVIDUAL SAMPLING NOISE

$$D_j = \beta_1 + d_j + \eta_j$$

$$\eta_j = \bar{\varepsilon}_{1j} - \bar{\varepsilon}_{0j}$$

For this fixed center, $\beta_1 + d_j$ remains constant across repeated samples of patients.

Only the individual errors contribute to the sampling variability of $D_j$.

### APPLY THE VARIANCE RULES

Let n be the number of patients per treatment per center; here n = 10.

$$\mathrm{Var}(\bar{\varepsilon}_{1j}) = \mathrm{Var}(\bar{\varepsilon}_{0j}) = \frac{\sigma^2}{n}$$

The two groups have independent errors, so the variance of their difference is:

$$\mathrm{Var}(\eta_j) = \frac{\sigma^2}{n} + \frac{\sigma^2}{n} = \frac{2\sigma^2}{n}$$

Differences between the centers’ true treatment effects do not add sampling variation within a fixed center.

# Uncertainty in the average across fixed centers

Average the J independent center comparisons. The fixed treatment deviations sum to zero.

$$\widehat{\beta}_1 = \beta_1 + \frac{1}{J}\sum_{j=1}^{J}\eta_j$$

$$\mathrm{Var}(\widehat{\beta}_1) = \frac{1}{J^2}\sum_{j=1}^{J}\frac{2\sigma^2}{n} = \frac{2\sigma^2}{Jn}$$

The individual errors are the only source of sampling variation around the true average for these centers.

More patients in these same centers can make this average increasingly precise.

Conditioning on the included centers, treatment heterogeneity adds no extra variance term for their average.

# Estimate the standard error from individual variation

### FROM THE MODEL RESIDUALS

The model fits a separate mean for every treatment–center combination.

Residual standard deviation: 25.66 m, estimating the individual-error standard deviation σ.

$$\widehat{\mathrm{SE}}(\widehat{\beta}_1) = \widehat{\sigma}\sqrt{\frac{2}{Jn}}$$

Use J = 6 centers and n = 10 patients per treatment per center.

### THE AVERAGE TREATMENT COMPARISON

**Estimated difference: 21.27 m**

Estimated standard error: 4.68 m

95% confidence interval:
11.98 to 30.55 m

The t interval uses 108 residual degrees of freedom: 120 patients minus 12 fitted cell means.

This interval describes uncertainty about the true average treatment effect in the six included centers.

# What if we omit the fixed interaction?

| Fixed-effects model | Estimate (m) | Estimated SE (m) |
|---|---|---|
| **Treatment + center** | 21.27 | 5.17 |
| **Treatment × center** | 21.27 | 4.68 |

Exact balance keeps the estimate equal to the average of the six observed treatment differences.

Omitting the interaction leaves systematic treatment-by-center differences in the residuals.

This inflates the estimated residual variance in expectation, and therefore the usual estimated variance of the average.

Here the estimated standard error changes; the average estimator and its actual sampling variance do not.

*PART 3*

# From included centers to a population

# What changes if the study uses different centers?

### THESE SIX CENTERS

Repeat the study with new patients in the same six centers.

Each center’s true treatment effect stays the same.

Uncertainty comes from estimating those six effects using patient outcomes.

### A POPULATION OF CENTERS

Repeat the study with new centers sampled from a wider population.

The selected centers can have different true treatment effects.

Uncertainty also comes from which centers enter the study.

A population-average treatment effect requires accounting for variation in treatment effects between sampled centers.

# Random treatment effects add a source of uncertainty

Let $\beta_1$ be the population-average treatment effect, with independent random center deviations:

$$d_j \overset{\text{iid}}{\sim} N(0, \sigma_d^2)$$

The deviations have expectation zero; their realized average need not be zero.

$$\widehat{\beta}_1 = \beta_1 + \frac{1}{J}\sum_{j=1}^{J} d_j + \frac{1}{J}\sum_{j=1}^{J} \eta_j$$

Assume the random center effects are independent of the individual errors. Then:

$$\text{Var}(\widehat{\beta}_1) = \frac{\sigma_d^2}{J} + \frac{2\sigma^2}{Jn}$$

The two variance terms reflect center sampling and individual sampling noise.

# More patients and more centers answer different needs

### AVERAGE IN THE INCLUDED CENTERS

Increasing patients per treatment makes each center's effect more precise.

$$\frac{2\sigma^2}{Jn} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty$$

Even heterogeneous treatment effects can have a precisely estimated average within these centers.

### AVERAGE IN THE POPULATION

More patients reduce individual noise, but the study still samples only J centers.

$$\frac{\sigma_d^2}{J} + \frac{2\sigma^2}{Jn} \longrightarrow \frac{\sigma_d^2}{J}$$

More centers reduce both variance terms. Six precisely measured centers still provide only six center effects.

These are sampling-variance formulas under the balanced model; fitted uncertainty also involves estimating the variance components.

# The target determines which uncertainty belongs

### FIXED INTERACTION

**Target: the average effect in the included centers.**

Patient sampling determines uncertainty around that fixed average.

Omitting the interaction can inflate the estimated uncertainty by leaving heterogeneity in the residuals.

### RANDOM TREATMENT EFFECT

**Target: the average effect in a population of centers.**

Center sampling adds uncertainty about that population average.

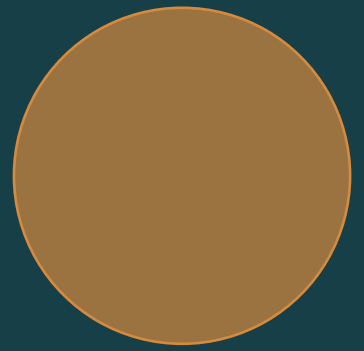
A random-intercept-only model can understate uncertainty because it omits treatment-effect variation.

Next: model a population of center-specific intercepts and treatment effects using random intercepts and slopes.

*BACKUP*

# Further discussion





# Why are the estimated standard errors almost identical?

### MOST PRECISION COMES FROM WITHIN CENTERS

Both models compare enhanced and standard care within each center.

These comparisons are informative because the treatment groups share the same center baseline. Baseline differences between centers do not obscure these comparisons.
Both models obtain most of their precision from this same source.

### BETWEEN CENTERS: LITTLE EXTRA PRECISION HERE

The random-intercept model can also use average outcomes from centers with different treatment proportions.

Centers differ substantially in baseline outcomes. A higher center average could reflect its baseline as well as its treatment allocation.

Under the random-intercept assumptions, this noisy evidence adds little precision: the estimated standard error barely decreases.

The additional evidence shifts the estimate slightly, but adds little precision: the estimated standard errors are 4.56 and 4.55 m.